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(4) Gradient Descent, Step-by-Ste | steepest descent example - Goog | lecture10.pdf | Convex function - Wikipedia | Definite matrix - Wikipedia

en.wikipedia.org/wiki/Definite_matrix#:~:text=A%20matrix%20that%20is%20not,it%20defines%20an%20inner%20product.

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example

$$N = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

for which $\begin{bmatrix} -1 & 1 \end{bmatrix} N \begin{bmatrix} -1 & 1 \end{bmatrix}^{\mathsf{T}} = -2 < 0.$

det

$|H - \lambda I| = 0$

$\lambda \to$ eigenvalues

Eigenvalues [edit]

Let M be an $n \times n$ Hermitian matrix (this includes real symmetric matrices). All eigenvalues of M are real, and their sign characterize its definiteness:

- M is positive definite if and only if all of its eigenvalues are positive. $\Leftrightarrow$ > 0
- M is positive semi-definite if and only if all of its eigenvalues are non-negative. $\to \geq 0$ $|H - \lambda I| = 0$
- M is negative definite if and only if all of its eigenvalues are negative. $\to < 0$
- M is negative semi-definite if and only if all of its eigenvalues are non-positive. ≤ 0
- M is indefinite if and only if it has both positive and negative eigenvalues.

Let PDP^{-1} be an eigendecomposition of M, where P is a unitary complex matrix whose columns comprise an orthonormal basis of eigenvectors of M, and D is a *real* diagonal m diagonal contains the corresponding eigenvalues. The matrix M may be regarded as a diagonal matrix D that has been re-expressed in coordinates of the (eigenvectors) basis P. P to some vector $\mathbf{z}$, giving $M\mathbf{z}$, is the same as changing the basis to the eigenvector coordinate system using P^{-1}, giving $P^{-1}\mathbf{z}$, applying the stretching transformation D to the result, gi changing the basis back P, giving $PDP^{-1}\mathbf{z}$.

With this in mind, the one-to-one change of variable $\mathbf{y} = P\mathbf{z}$ shows that $\mathbf{z}^* M\mathbf{z}$ is real and positive for any complex vector $\mathbf{z}$ if and only if $\mathbf{y}^* D\mathbf{y}$ is real and positive for any y; in othe definite. For a diagonal matrix, this is true only if each element of the main diagonal—that is, every eigenvalue of M—is positive. Since the spectral theorem guarantees all eigenvalu to be real, the positivity of eigenvalues can be checked using Descartes' rule of alternating signs when the characteristic polynomial of a real, symmetric matrix M is available.

Decomposition [edit]

example for steep....pdf | Show all

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